

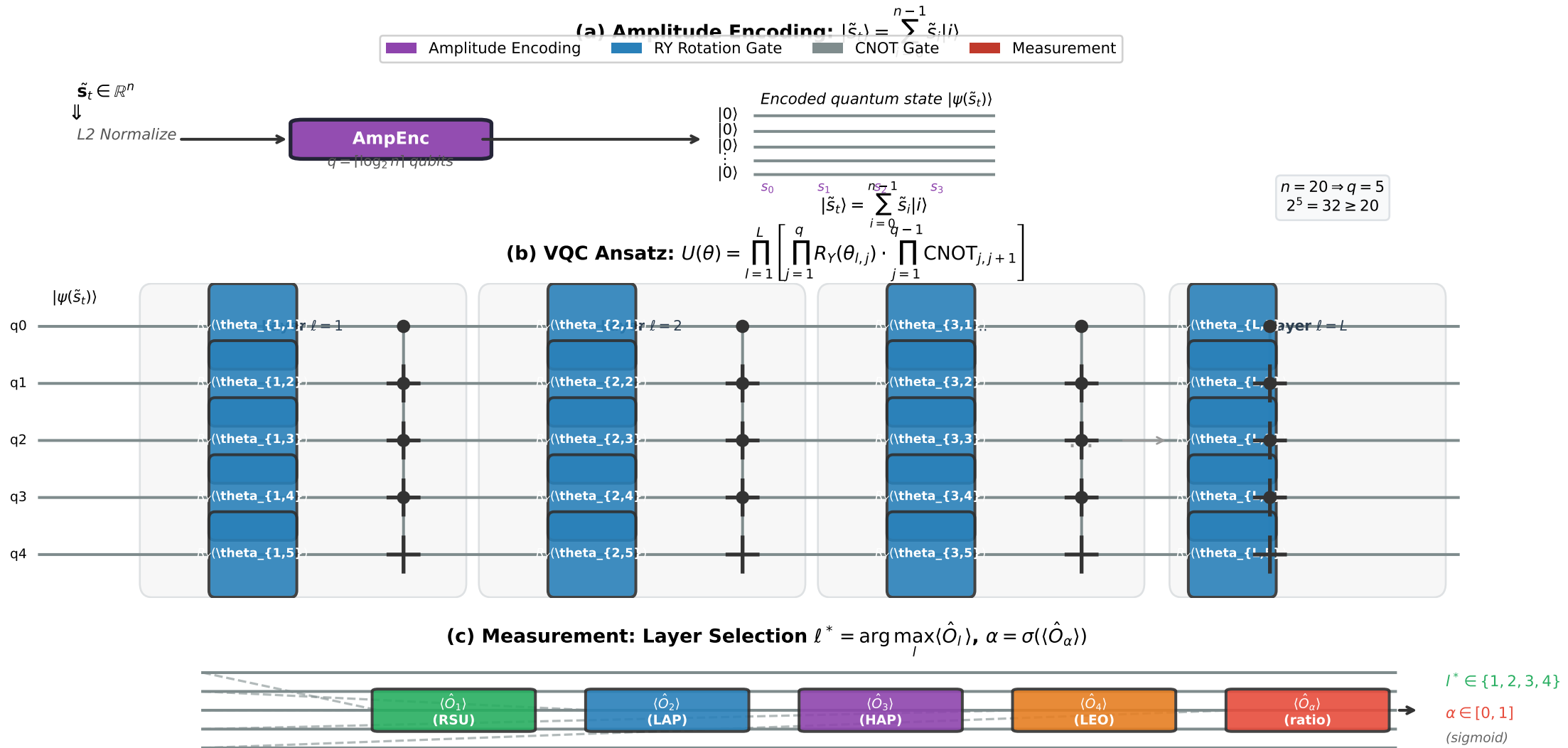


Fig QC-1: Quantum-HRL VQC architecture. (a) Amplitude Encoding maps the n -dimensional state vector onto $q = \lceil \log_2 n \rceil$ qubits. (b) Each VQC layer applies parameterized R_Y rotations followed by a CNOT entangling ladder. (c) Measurement of Pauli observables yields layer selection ℓ^* and offloading ratio α .